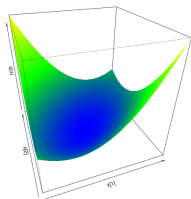
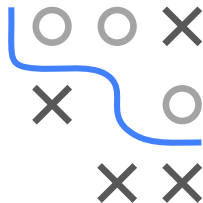


Optimization in Machine Learning

Foundations

Quadratic functions



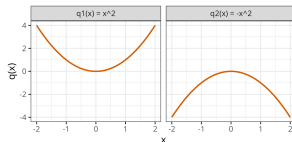
Learning goals

- Definition of quadratic functions
- Gradient, Hessian
- Optima
- Geometry: spectrum and curvature
- Condition number

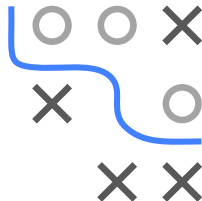
UNIVARIATE QUADRATIC

- Quadratic function $q : \mathbb{R} \rightarrow \mathbb{R}$

$$q(x) = ax^2 + bx + c, \quad a \neq 0$$

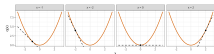


- Left: $q_1(x) = x^2$. Right: $q_2(x) = -x^2$

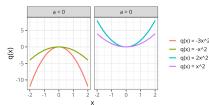


UNIVARIATE: BASIC PROPERTIES

- Slope at $(x, q(x))$:



- Curvature:



- $a > 0$: q convex, bounded from below, unique global minimum
- $a < 0$: q concave, bounded from above, unique global maximum
- Optimum x^*

$$q'(x) = 0 \Leftrightarrow 2ax + b = 0 \Rightarrow x^* = \frac{-b}{2a}$$

as 2nd derivative: $q''(x^*) = 2a \neq 0$

SYMMETRIZATION

- W.l.o.g. assume $\mathbf{A}$ symmetric, i.e., $\mathbf{A}^T = \mathbf{A}$
- If $\mathbf{A}$ not symmetric, there exists symmetric $\tilde{\mathbf{A}}$ with

$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{x}^T \tilde{\mathbf{A}} \mathbf{x} =: \tilde{q}(\mathbf{x})$$

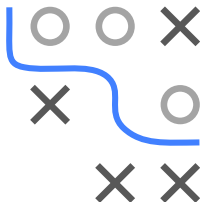
- Justification

$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \frac{1}{2} \mathbf{x}^T \underbrace{(\mathbf{A} + \mathbf{A}^T)}_{\tilde{\mathbf{A}}_1} \mathbf{x} + \frac{1}{2} \mathbf{x}^T \underbrace{(\mathbf{A} - \mathbf{A}^T)}_{\tilde{\mathbf{A}}_2} \mathbf{x}$$

- $\tilde{\mathbf{A}}_1$ symmetric, $\tilde{\mathbf{A}}_2$ anti-symmetric ($\tilde{\mathbf{A}}_2^T = -\tilde{\mathbf{A}}_2$)
- Since $\mathbf{x}^T \mathbf{A}^T \mathbf{x}$ is a scalar, equal to its transpose

$$\begin{aligned} \mathbf{x}^T (\mathbf{A} - \mathbf{A}^T) \mathbf{x} &= \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{x}^T \mathbf{A}^T \mathbf{x} = \mathbf{x}^T \mathbf{A} \mathbf{x} - (\mathbf{x}^T \mathbf{A}^T \mathbf{x})^T \\ &= \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{x}^T \mathbf{A} \mathbf{x} = 0 \end{aligned}$$

- Therefore $\tilde{q}(\mathbf{x}) = \mathbf{x}^T \tilde{\mathbf{A}} \mathbf{x}$ with $\tilde{\mathbf{A}} = \tilde{\mathbf{A}}_1/2$



GRADIENT AND HESSIAN

- $q: \mathbb{R}^d \rightarrow \mathbb{R}$

$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

- Gradient

$$\nabla q(\mathbf{x}) = ((\mathbf{A}^T + \mathbf{A})\mathbf{x} + \mathbf{b})^T$$

- Under assumed symmetry: $\nabla q(\mathbf{x}) = (2\mathbf{A}\mathbf{x} + \mathbf{b})^T$

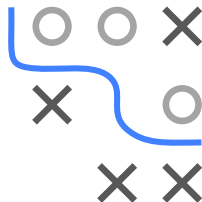
- Directional derivative: $\nabla q(\mathbf{x}) \mathbf{v}$

- Hessian

$$\nabla^2 q(\mathbf{x}) = (\mathbf{A}^T + \mathbf{A}) = 2\mathbf{A} =: \mathbf{H} \in \mathbb{R}^{d \times d}$$

- Under assumed symmetry: $\mathbf{H} = 2\mathbf{A}$

- Directional curvature: $\mathbf{v}^T \mathbf{H} \mathbf{v}$



OPTIMUM

- $q: \mathbb{R}^d \rightarrow \mathbb{R}$

$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

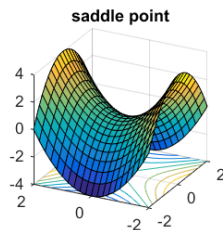
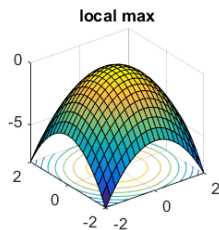
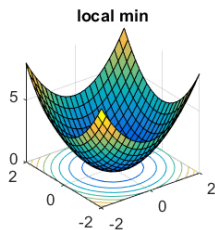
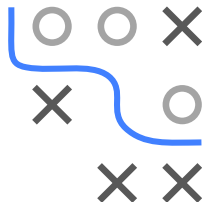
- Since $\mathbf{A}$ full rank, unique stationary point $\mathbf{x}^*$ (min, max, or saddle)

$$\nabla q(\mathbf{x}^*) = \mathbf{0}^T$$

$$(2\mathbf{A}\mathbf{x}^* + \mathbf{b})^T = \mathbf{0}^T$$

$$\mathbf{x}^* = -\frac{1}{2}\mathbf{A}^{-1}\mathbf{b}$$

- $q(\mathbf{x}^*) = c - \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$



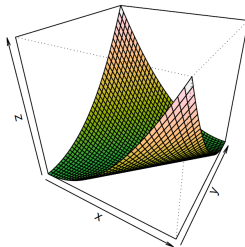
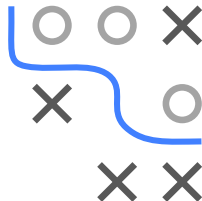
- Left: $\mathbf{A}$ pos. def. Middle: $\mathbf{A}$ neg. def. Right: $\mathbf{A}$ indefinite

OPTIMA: RANK-DEFICIENT CASE

- $q : \mathbb{R}^d \rightarrow \mathbb{R}$

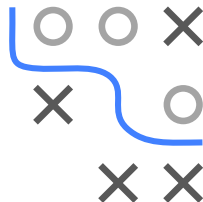
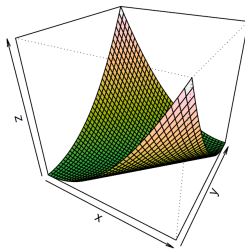
$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

- Assume $\mathbf{A}$ symmetric now
- For stationary points to exist, we need : $\nabla q(\mathbf{x}) = 2\mathbf{A}\mathbf{x} + \mathbf{b} = 0$
- This implies we need $\mathbf{b} \in \text{range}(\mathbf{A})$, let's assume this is the case
- Let $\mathbf{x}_p$ be stationary, so $2\mathbf{A}\mathbf{x}_p = -\mathbf{b}$
- Then any point in affine space $\mathbf{x}_p + \ker(\mathbf{A})$ is also stationary, with same function value and same Hessian (as it is constant)



OPTIMA: RANK-DEFICIENT CASE

- All affine spaces of form $\mathbf{x}_p + \ker(\mathbf{A})$ for diff. valid $\mathbf{x}_p$ are the same
- Any stationary point must be in $\mathbf{x}_p + \ker(\mathbf{A})$
- So $\mathbf{x}_p + \ker(\mathbf{A})$ are all the stationary points, with same curvature
- If $\mathbf{A} \succeq 0$, they are all minima
- If $\mathbf{A} \preceq 0$, they are all maxima
- If $\mathbf{A}$ is indefinite, they are all saddle points

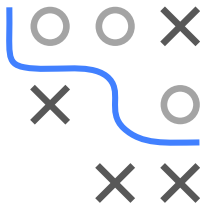
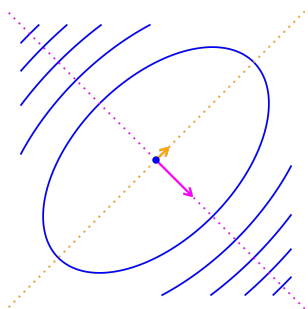


GEOMETRY OF QUADRATIC FUNCTIONS

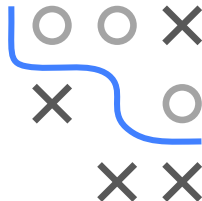
- Example: $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \Rightarrow \mathbf{H} = 2\mathbf{A} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$
- Since $\mathbf{H}$ symmetric: eigendecomposition $\mathbf{H} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$

$$\mathbf{V} = \begin{pmatrix} | & | \\ \textcolor{violet}{v}_{\max} & \textcolor{brown}{v}_{\min} \\ | & | \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \text{ orthogonal}$$

$$\mathbf{\Lambda} = \begin{pmatrix} \textcolor{violet}{\lambda}_{\max} & 0 \\ 0 & \textcolor{brown}{\lambda}_{\min} \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$$



SPECTRUM AND CURVATURE

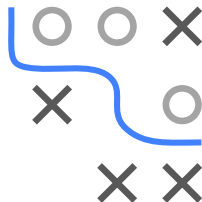


- $\mathbf{v}_{\max}$ direction of highest curvature, with curvature value $\lambda_{\max}$

$$\mathbf{v}^T \mathbf{H} \mathbf{v} = \mathbf{v}^T \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T \mathbf{v} = \mathbf{w}^T \mathbf{\Lambda} \mathbf{w} = \sum_{i=1}^d \lambda_i w_i^2 \leq \lambda_{\max} \sum_{i=1}^d w_i^2 = \lambda_{\max} \|\mathbf{w}\|^2$$

- Since $\|\mathbf{v}\| = \|\mathbf{x}\|$ ($\mathbf{V}$ orthogonal): $\max_{\|\mathbf{v}\|=1} \mathbf{v}^T \mathbf{H} \mathbf{v} \leq \lambda_{\max}$
- For $\mathbf{v}_{\max}$ we obtain this upper bound: $\mathbf{v}_{\max}^T \mathbf{H} \mathbf{v}_{\max} = \mathbf{e}_1^T \mathbf{\Lambda} \mathbf{e}_1 = \lambda_{\max}$
- Analogously, $\mathbf{v}_{\min}$ direction of lowest curvature, with curvature value $\lambda_{\min}$
- Contour lines of any quadratic function are ellipses

SECOND-ORDER SUFFICIENT CONDITION



- Recall: Second order condition for optimality is sufficient
- If $H(\mathbf{x}^*) \succ 0$ at stationary point $\mathbf{x}^*$, then $\mathbf{x}^*$ local minimum ($\prec$ for maximum)

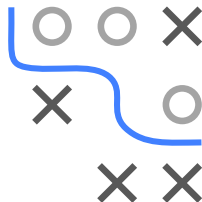
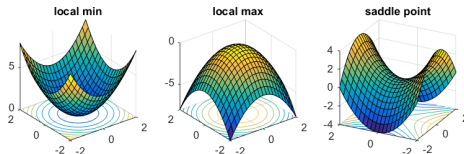
$$f(\mathbf{x}) = f(\mathbf{x}^*) + \underbrace{\nabla f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*)}_{=0} + \underbrace{\frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T H(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*)}_{\geq \lambda_{\min} \|\mathbf{x} - \mathbf{x}^*\|^2} + \underbrace{R_2(\mathbf{x}, \mathbf{x}^*)}_{=o(\|\mathbf{x} - \mathbf{x}^*\|^2)}$$

- Choose $\epsilon > 0$ s.t. $|R_2(\mathbf{x}, \mathbf{x}^*)| < \frac{1}{2} \lambda_{\min} \|\mathbf{x} - \mathbf{x}^*\|^2$ for $\mathbf{x} \neq \mathbf{x}^*$, $\|\mathbf{x} - \mathbf{x}^*\| < \epsilon$

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + \underbrace{\frac{1}{2} \lambda_{\min} \|\mathbf{x} - \mathbf{x}^*\|^2}_{>0} + R_2(\mathbf{x}, \mathbf{x}^*) > f(\mathbf{x}^*)$$

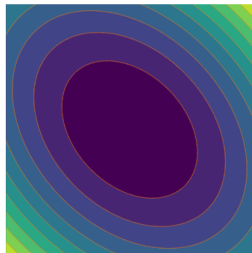
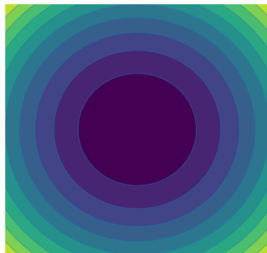
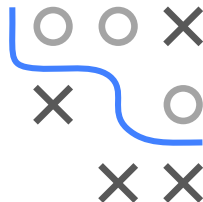
EIGENVALUES AND SHAPE

- If spectrum of $\mathbf{A}$ is known, also that of $\mathbf{H} = 2\mathbf{A}$ is known
- If all eigenvalues of $\mathbf{H} \stackrel{(>)}{\geq} 0 \Leftrightarrow \mathbf{H} \stackrel{(>)}{\succ} 0$:
 - q (strictly) convex
 - (Unique) global minimum
- If all eigenvalues of $\mathbf{H} \stackrel{(<)}{\leq} 0 \Leftrightarrow \mathbf{H} \stackrel{(<)}{\preceq} 0$:
 - q (strictly) concave
 - (Unique) global maximum
- If $\mathbf{H}$ has both positive and negative eigenvalues ($\Leftrightarrow \mathbf{H}$ indefinite):
 - q neither convex nor concave
 - there is a saddle point



CONDITION AND CURVATURE

- $\kappa(\mathbf{H}) = \kappa(\mathbf{A}) = |\lambda_{\max}|/|\lambda_{\min}|$
- High condition means
 - $|\lambda_{\max}| \gg |\lambda_{\min}|$
 - Curvature along $\mathbf{v}_{\max} \gg$ along $\mathbf{v}_{\min}$
 - Problem for algorithms like gradient descent

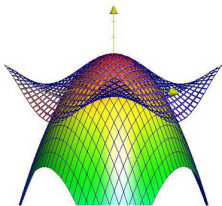


Left: Excellent condition. Middle: Good condition. Right: Bad condition.

APPROXIMATION OF SMOOTH FUNCTIONS

- Any $f \in \mathcal{C}^2$ can be locally approximated by quadratic function (second order Taylor)

$$f(\mathbf{x}) \approx f(\tilde{\mathbf{x}}) + \nabla f(\tilde{\mathbf{x}})(\mathbf{x} - \tilde{\mathbf{x}}) + \frac{1}{2}(\mathbf{x} - \tilde{\mathbf{x}})^T \nabla^2 f(\tilde{\mathbf{x}})(\mathbf{x} - \tilde{\mathbf{x}})$$



► [Click for source](#)

f and second order approximation: dark vs bright grid.

- $\implies$ Hessians provide information about **local** geometry of a function

